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EDUCATION

Berea College B.S. in Physics and Mathematics (double major), minor in Computer Science. Dean’s List every term.	May 2028 Berea, KY
Lyceum of Belarusian State University STEM magnet high school, physics track.	2022 – 2024 Minsk, Belarus

PUBLICATIONS AND PREPRINTS

[1] **A. Melnichenka***, P. Liong*, A. Bukhtatyi, A. Bilous, L. Levitov. “Spontaneous Running Waves and Self-Oscillatory Transport in Dirac Fluids.” *Under review at Physical Review Letters*, arXiv:2512.16571, 2025. *Equal contribution.

[2] **A. Melnichenka**, A. Lazarian, D. Pogosyan. “Recovering 3D Magnetic Turbulence from a Single Polarization Map.” *Submitted to The Astrophysical Journal*, arXiv:2602.22204, 2026.

[3] **A. Melnichenka**, X. Shen, V. Tran, M. Vogelsberger. “Drift Diffusion Coefficients for Self-Interacting Dark Matter.” *In preparation*, 2026.

RESEARCH EXPERIENCE

Electron-Fluid Instabilities in Dirac Materials <i>Undergraduate Researcher with Prof. Leonid Levitov</i> First author on a manuscript under review at <i>Physical Review Letters</i> on current-driven self-sustained waves and self-oscillatory transport in Dirac electron fluids. Developed the full computational pipeline: pseudospectral PDE solver, collision-operator eigenmode computations on large grids, instability-threshold mapping, and wavelength-selection analysis across multiple initial conditions. Produced the simulations and core numerical evidence for the project; work was presented at the 2026 APS Global Physics Summit by Prof. Leonid Levitov.	MIT, Condensed Matter Theory 2025 – present
Turbulence Recovery from Radio Polarization <i>Undergraduate Researcher with Prof. Alex Lazarian and Prof. Dmitri Pogosyan</i> Developed a single-frequency polarization diagnostic that recovers 3D inertial-range magnetic turbulence statistics from one radio map. First-author manuscript submitted to <i>The Astrophysical Journal</i> . Validated on synthetic Faraday screens and AthenaK MHD turbulence simulations in sub-Alfvénic and super-Alfvénic regimes. Built a reproducible pipeline with checks for interferometric filtering, ready for LOFAR, MeerKAT, and VLA data.	UW-Madison, Astronomy 2025 – present
Self-Interacting Dark Matter near Supermassive Black Holes <i>Undergraduate Researcher with Prof. Mark Vogelsberger (mentor: Xuejian Shen)</i> Reproduced published benchmarks for SIDM evolution near supermassive black holes and isolated discrepancies via targeted ablation tests. Developing an orbit-averaged stochastic loss-cone model with scaling predictions for collapse timescales.	MIT Kavli Institute for Astrophysics 2025 – present
Caltech SURF Research Program <i>Summer Research Fellow with Prof. Jim Fuller</i> Selected for the 2026 Caltech SURF program to conduct theoretical astrophysics research with Prof. Jim Fuller.	California Institute of Technology Summer 2026

SELECTED TALKS

American Astronomical Society, 247th Meeting <i>Oral presentation on single-frequency Faraday-screen tomography</i>	Phoenix, AZ Jan 2026
American Physical Society, Division of Plasma Physics <i>Contributed oral talk on 3D magnetic turbulence recovery from polarization maps</i>	Long Beach, CA Nov 2025
The Magnetized Turbulent Universe (Conference Honoring A. Lazarian) <i>Invited talk on polarization-angle statistics and crossover scaling. Only undergraduate speaker.</i>	Playa del Carmen, Mexico Nov 2025

HONORS, FELLOWSHIPS, AND SELECTED PROGRAMS

Emergent Ventures Fellowship (\$23,000), Mercatus Center at George Mason University, selected by Tyler Cowen, 2026.

Sigma Pi Sigma, national physics honor society; first sophomore in department’s 96-year history inducted, 2026.

Caltech SURF Research Fellow, selected to work with Prof. Jim Fuller, 2026.

International Theoretical Physics Practicum, returning participant, 2026.

Putnam Mathematical Competition, score 28, top 13% nationally, 2025.

Belarus National Physics Olympiad: Gold medal (2022), Silver medal (2024), Bronze medal (2023).

IPhO Reserve Training Camp, top six nationally in Belarus, 2022.

First Degree Diploma, Phystech.International (Moscow Institute of Physics and Technology), 2023.

Presidential Award for Gifted Youth (Belarus), 2022.

PROJECTS, LEADERSHIP, AND OUTREACH

Trajecta Founder
Publishing platform for serious technical writing 2026 – present
Building a platform where the right to publish is earned through demonstrated competence. Designing the architecture, contributor-evaluation model, editorial standards, and publishing workflow.
Awarded a \$23,000 Emergent Ventures fellowship from the Mercatus Center at George Mason University, selected by Tyler Cowen.

BelSO.org Founder and Lead Engineer
National problem archive and training platform for Belarusian science olympiads 2023 – present
Built a unified platform spanning physics, mathematics, astronomy, and informatics, with 8,500 problems, 3,200 solutions, and roughly 12,500 monthly users.
Used by olympiad students and coaches across Belarus; recognized on International Physics Olympiad country resource pages.

SavchenkoSolutions.com Founder
Collaborative solution platform for Savchenko’s Problems in Physics 2023 – present
Built a community-maintained platform around a canonical advanced problem collection: 67 contributors from 40 countries, 1,493 published solutions, and roughly 1,500 daily users.
Personally authored 847 solutions and produced the first English translations for 362 problems; designed the editorial workflow, review queue, and bilingual publication standards.

SOFTWARE AND TECHNICAL SKILLS

Languages and Tools Python (NumPy, SciPy, Astropy, JAX), C, Bash, Git, L^AT_EX
Methods Spectral PDE solvers, turbulence statistics, Monte Carlo and Langevin methods, polarization correlation analysis, structure functions, numerical ODEs and PDEs
Open-Source Packages AstroTurbulence (polarization statistics), electronic-kapitsa-waves (spectral synthesis), SIDM_Transport_Theory_vs_MC (drift diffusion benchmarks). All with commit-pinned figures and environment specifications.
Human Languages English (fluent), Belarusian (native), Russian (native)

TEACHING

Berea College, Physics Department Berea, KY
Teaching Assistant Jan 2025 – present
Led weekly office hours and review sessions for introductory and calculus-based physics. Coached students on scientific writing, error checking, and reproducible computational workflows.